MATH 231 – Linear Algebra I

The Matrix Unit, Part 4 ▪ Area, and the Rotation Matrix

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About this document. This is Part 4 of the matrix unit, Sections 22–25. Section numbers run continuously through the unit (Part 2's opening table lists them all). References into the vector unit always name it.

Assumed. The 2×2 determinant and its meaning (§14); the gallery of motions and the two families of 2×2 orthogonal matrices (§16–§21); the quarter-turn $\mathbf{J}$ (§19.1); the laws M1–M6 (§8.4). From the vector unit: the angle between two vectors (§10.3), the dot product formula $\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$ (§11.3), and the Cauchy–Schwarz inequality (§13).

What you should be able to do afterwards. Use the properties of the 2×2 determinant to compute determinants of scaled, transposed and multiplied matrices without multiplying them out; derive the area of a parallelogram from base and height and show it equals $|\det \mathbf{A}|$; derive the rotation matrix $\mathbf{R}_\theta$ from the unit circle, and show it is orthogonal with determinant 1; compose and undo rotations; and write the matrices of rotations of $\mathbb{R}^3$ about the coordinate axes.

22. Properties of the 2×2 determinant

Throughout this section

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}, \qquad \det \mathbf{A} = ad - bc, \qquad \det \mathbf{B} = eh - fg.$$

Each property comes with a one-line proof by computation, and each is worth reading against the gallery of Part 3. The same properties hold for $n \times n$ matrices, in the form Part 7 gives them.

22.1. The identity and the transpose

P1. $\det \mathbf{I} = 1 \cdot 1 - 0 \cdot 0 = 1$. The identity moves nothing, so every area is unchanged.

P2. $\det \mathbf{A}^\mathsf{T} = \det \mathbf{A}$. Indeed $\mathbf{A}^\mathsf{T} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ has determinant $ad - cb$. This is the observation of §12.4 that the same number tests the rows and the columns. Every statement below about columns is therefore true of rows too.

22.2. Scaling

P3. $\det(\alpha\mathbf{A}) = \alpha^2 \det \mathbf{A}$. Indeed $(\alpha a)(\alpha d) - (\alpha b)(\alpha c) = \alpha^2(ad - bc)$.
Not $\alpha \det \mathbf{A}$: scaling the matrix scales *both* columns, so both sides of the parallelogram, and the area picks up the factor twice. With $\mathbf{A} = \mathbf{I}$ this is the uniform scaling of §16.1: $\det(\alpha\mathbf{I}) = \alpha^2$.

22.3. The columns

P4. *Swapping the two columns changes the sign.* $\det \begin{bmatrix} b & a \\ d & c \end{bmatrix} = bc - ad = -\det \mathbf{A}$. By P2, the same holds for swapping the rows. (This is the pair of matrices in the middle and left pictures of §14.4.)

P5. *The determinant is linear in each column separately.* Write $\det[\,\mathbf{u}\ \ \mathbf{w}\,]$ for the determinant of the matrix with columns $\mathbf{u}$ and $\mathbf{w}$. Then

$$\det[\,\alpha\mathbf{u}+\beta\mathbf{v}\ \ \mathbf{w}\,] = \alpha\det[\,\mathbf{u}\ \ \mathbf{w}\,] + \beta\det[\,\mathbf{v}\ \ \mathbf{w}\,],$$

since $(\alpha u_1+\beta v_1)w_2-(\alpha u_2+\beta v_2)w_1 = \alpha(u_1w_2-u_2w_1)+\beta(v_1w_2-v_2w_1)$. Likewise in the second column.

A common confusion (Linear in each column is not linear in the matrix). P5 holds one column fixed. The determinant of a sum of matrices is *not* the sum of the determinants: $\det(\mathbf{I}+\mathbf{I}) = \det(2\mathbf{I}) = 4$, while $\det\mathbf{I}+\det\mathbf{I} = 2$.

P6. *Two parallel columns give zero.* If $\mathbf{w} = k\mathbf{u}$ then $u_1(ku_2)-u_2(ku_1) = 0$. In particular two equal columns give zero.

P7. *Adding a multiple of one column to the other changes nothing.* By P5 and P6,

$$\det[\,\mathbf{a}_1\ \ \mathbf{a}_2+k\mathbf{a}_1\,] = \det[\,\mathbf{a}_1\ \ \mathbf{a}_2\,] + k\det[\,\mathbf{a}_1\ \ \mathbf{a}_1\,] = \det\mathbf{A}+0.$$

Geometrically, $\mathbf{a}_2$ slides parallel to $\mathbf{a}_1$, which is a shear of the parallelogram (§18): the base and the height stay the same, so the area does too.

22.4. Products

P8. $\det(\mathbf{AB}) = \det\mathbf{A}\,\det\mathbf{B}$. The product is $\mathbf{AB} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$, so

$$\det(\mathbf{AB}) = (ae+bg)(cf+dh)-(af+bh)(ce+dg).$$

Expanding gives eight terms. The four $aecf-afce$ and $bgdh-bhdg$ cancel in pairs, and the rest regroup:

$$aedh+bgcf-afdg-bhce = ad(eh-fg)-bc(eh-fg) = (ad-bc)(eh-fg).$$

Two consequences are worth stating. First, by §15.4 the product is "first $\mathbf{B}$, then $\mathbf{A}$", and P8 says the area factors multiply: $\mathbf{B}$ scales areas by $|\det\mathbf{B}|$, then $\mathbf{A}$ by $|\det\mathbf{A}|$. Second, $\det(\mathbf{AB}) = \det(\mathbf{BA})$, because both equal the product of two numbers. The matrices $\mathbf{AB}$ and $\mathbf{BA}$ are usually different (§8.4), and their determinants are always equal, like their traces (§9.3).

22.5. Special matrices

P9. *Triangular and diagonal.* $\det\begin{bmatrix} a & b \\ 0 & d \end{bmatrix} = ad$, $\det\begin{bmatrix} a & 0 \\ c & d \end{bmatrix} = ad$, and $\det\operatorname{diag}(s_1,s_2) = s_1s_2$: the product of the diagonal.

P10. *Orthogonal.* If $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$ then by P8, P2 and P1, $(\det\mathbf{Q})^2 = \det\mathbf{Q}^{\mathsf{T}}\det\mathbf{Q} = \det(\mathbf{Q}^{\mathsf{T}}\mathbf{Q}) = \det\mathbf{I} = 1$, so $\det\mathbf{Q} = \pm1$. This is what §21.2 found by listing every case, now obtained without the list.

P11. $\det\mathbf{A} \neq 0$ exactly when $\operatorname{rank}\mathbf{A} = 2$ (§14.2).

22.6. The gallery, by determinant

Motion	det	**Which property explains it**
uniform scaling $\alpha\mathbf{I}$	α^2	P3 with $\mathbf{A} = \mathbf{I}$
scaling the axes, $\mathrm{diag}(s_1, s_2)$	$s_1 s_2$	P9
shear $\left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$	1	P9, or P7 applied to $\mathbf{I}$
rotation	1	P10, the $+1$ family
reflection	-1	P10, the -1 family; or P4 applied to $\mathbf{I}$ for $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$
projection $\mathrm{diag}(1, 0)$	0	P9, or P11 (rank 1)

23. The determinant is an area

§14.3 claimed that the parallelogram with sides $\mathbf{a}_1$ and $\mathbf{a}_2$ has area $|\det \mathbf{A}|$. This section proves it, starting from the oldest area formula there is.

23.1. Base times height

Let the columns of $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be

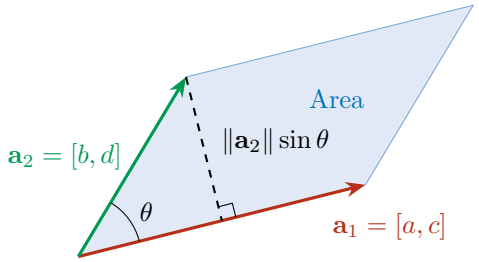
$$\mathbf{a}_1 = \begin{bmatrix} a \\ c \end{bmatrix}, \qquad \mathbf{a}_2 = \begin{bmatrix} b \\ d \end{bmatrix},$$

linearly independent, so that neither is zero and they are not parallel. Let θ be the angle between them (the vector unit, §10.3); then $0 < \theta < \pi$. The area of a parallelogram is its base times its height.

- Take the side $\mathbf{a}_1$ as the base. Its length is $\|\mathbf{a}_1\|$.
- The height is the distance from the tip of $\mathbf{a}_2$ to the line through $\mathbf{a}_1$. It is the side opposite the angle θ in a right triangle with hypotenuse $\|\mathbf{a}_2\|$, so it is $\|\mathbf{a}_2\| \sin\theta$.

So

$$\text{Area} = \|\mathbf{a}_1\| \, \|\mathbf{a}_2\| \, \sin\theta.$$



23.2. Getting rid of the sine

The sine is the awkward part: the entries a, b, c, d do not obviously know anything about $\sin\theta$. They do know about $\cos\theta$, through the dot product. So square the area, and trade $\sin^2\theta$ for $1 - \cos^2\theta$:

$$\text{Area}^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 \sin^2\theta = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 \big(1 - \cos^2\theta\big) = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \big(\|\mathbf{a}_1\| \|\mathbf{a}_2\| \cos\theta\big)^2.$$

From the vector unit. The last bracket is the dot product: $\mathbf{a}_1 \cdot \mathbf{a}_2 = \|\mathbf{a}_1\|\|\mathbf{a}_2\| \cos\theta$ (§11.3). So

$$\text{Area}^2 = \|\mathbf{a}_1\|^2\|\mathbf{a}_2\|^2 - (\mathbf{a}_1 \cdot \mathbf{a}_2)^2,$$

and every quantity on the right is computed from the entries alone.

23.3. In coordinates

With $\|\mathbf{a}_1\|^2 = a^2 + c^2$, $\|\mathbf{a}_2\|^2 = b^2 + d^2$ and $\mathbf{a}_1 \cdot \mathbf{a}_2 = ab + cd$,

$$\text{Area}^2 = (a^2 + c^2)(b^2 + d^2) - (ab + cd)^2.$$

Expand both products:

$$(a^2b^2 + a^2d^2 + c^2b^2 + c^2d^2) - (a^2b^2 + 2abcd + c^2d^2) = a^2d^2 - 2abcd + b^2c^2 = (ad - bc)^2.$$

23.4. The result

Take square roots. The area is not negative, and neither is $\sin\theta$ for $0 \le \theta \le \pi$, so

$$\text{Area} = \sqrt{(ad - bc)^2} = |ad - bc|.$$

Theorem 23.1. *The parallelogram with sides* $\mathbf{a}_1 = [\,a, c\,]$ *and* $\mathbf{a}_2 = [\,b, d\,]$ *has area*

$$\boxed{\text{Area} = |ad - bc| = |\det \mathbf{A}|} \qquad \text{where } \mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The formula also covers the case set aside at the start. If $\mathbf{a}_1$ and $\mathbf{a}_2$ are parallel, the parallelogram is flat, its area is 0, and so is $\det \mathbf{A}$ (P6).

Example 23.1. For $\mathbf{a}_1 = [\,3, 0\,]$ and $\mathbf{a}_2 = [\,1, 2\,]$, the base lies along the x-axis with length 3 and the height is 2, so the area is 6; and $\det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = 6$. For the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ of §13.4, base times height is less obvious, but the theorem gives $|4 - 1| = 3$ at once.

23.5. What else the identity says

Cauchy–Schwarz, with the gap measured. The middle line of §23.2 can be read as

$$\|\mathbf{a}_1\|^2\|\mathbf{a}_2\|^2 - (\mathbf{a}_1 \cdot \mathbf{a}_2)^2 = (\det \mathbf{A})^2 \;\ge\; 0.$$

That is the Cauchy–Schwarz inequality $|\mathbf{a}_1 \cdot \mathbf{a}_2| \le \|\mathbf{a}_1\|\|\mathbf{a}_2\|$ of the vector unit's §13, in $\mathbb{R}^2$, with the amount by which it is strict given exactly. Equality holds exactly when $\det \mathbf{A} = 0$, that is, when the vectors are parallel, which is the equality case found in the vector unit's §13.

A second formula for the area. The same quantity is the determinant of $\mathbf{A}^\mathsf{T}\mathbf{A}$:

$$\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} \|\mathbf{a}_1\|^2 & \mathbf{a}_1 \cdot \mathbf{a}_2 \\ \mathbf{a}_1 \cdot \mathbf{a}_2 & \|\mathbf{a}_2\|^2 \end{bmatrix}, \qquad \det(\mathbf{A}^\mathsf{T}\mathbf{A}) = \|\mathbf{a}_1\|^2\|\mathbf{a}_2\|^2 - (\mathbf{a}_1 \cdot \mathbf{a}_2)^2 = \text{Area}^2,$$

in agreement with P8 and P2: $\det(\mathbf{A}^\mathsf{T}\mathbf{A}) = (\det \mathbf{A})^2$.

23.6. Every area, not just one

A square of the grid with corner $\mathbf{p}$ and side h is spanned by $h\mathbf{e}_1$ and $h\mathbf{e}_2$. By linearity, $\mathbf{A}$ sends it to the parallelogram with corner $\mathbf{Ap}$ spanned by $h\mathbf{a}_1$ and $h\mathbf{a}_2$, whose area by the theorem and P3 is $|\det[\,h\mathbf{a}_1\ h\mathbf{a}_2\,]| = h^2|\det\mathbf{A}|$. Every small square has its area multiplied by the same factor $|\det\mathbf{A}|$. Any region can be approximated by small squares, and its image by the corresponding parallelograms, so

$$\text{area of the image of a region} = |\det\mathbf{A}| \times \text{area of the region.}$$

Read with this, P8 says that doing $\mathbf{B}$ and then $\mathbf{A}$ scales every area by $|\det\mathbf{B}|$ and then by $|\det\mathbf{A}|$. That is why the determinant of a product is the product of the determinants.

24. The rotation matrix

24.1. The unit circle, by angle

Every point of the unit circle can be written

$$[\,\cos t,\ \sin t\,]$$

for an angle t measured counterclockwise from the positive x-axis; that is what cosine and sine *are*. As t runs from 0 to 2π the point goes once around. The standard basis vectors sit at two particular angles: $\mathbf{e}_1 = [\,\cos 0, \sin 0\,]$ at $t = 0$, and $\mathbf{e}_2 = [\,\cos\frac{\pi}{2}, \sin\frac{\pi}{2}\,]$ at $t = \frac{\pi}{2}$.

24.2. Rotating by θ

Rotation by θ about the origin keeps every point at the same distance from the origin and adds θ to its angle. On the unit circle,

$$[\,\cos t,\ \sin t\,] \longmapsto [\,\cos(t+\theta),\ \sin(t+\theta)\,].$$

It is linear. A rotation moves the plane rigidly, so it carries the parallelogram with sides $\mathbf{x}$ and $\mathbf{y}$ to the parallelogram with sides the rotated $\mathbf{x}$ and the rotated $\mathbf{y}$. The diagonal goes with it, so the image of $\mathbf{x}+\mathbf{y}$ is the sum of the images. Stretching $\mathbf{x}$ before rotating is the same as stretching after. So §13.3 applies: the matrix is determined by where $\mathbf{e}_1$ and $\mathbf{e}_2$ go.

- $\mathbf{e}_1$, at angle 0, goes to the point at angle θ:

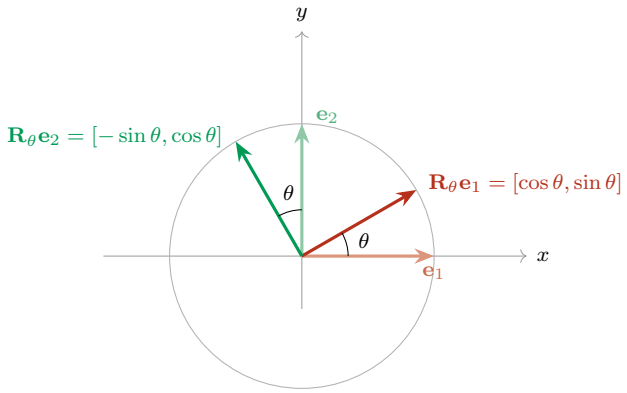
$$\mathbf{e}_1 \longmapsto [\,\cos\theta,\ \sin\theta\,].$$

- $\mathbf{e}_2$, at angle $\frac{\pi}{2}$, goes to the point at angle $\theta + \frac{\pi}{2}$. That point is the previous one turned a further quarter-turn, and §19.1 says a quarter-turn sends $[\,x, y\,]$ to $[\,-y, x\,]$:

$$\mathbf{e}_2 \longmapsto \mathbf{J}\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix} = [\,-\sin\theta,\ \cos\theta\,].$$

Definition 24.1 (Rotation matrix)**.** The matrix of the counterclockwise rotation of $\mathbb{R}^2$ by the angle θ is

$$\boxed{\mathbf{R}_\theta = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}.}$$



24.3. Checks

Example 24.1.

- $\theta = 0$: $\mathbf{R}_0 = \mathbf{I}$. No rotation.
- $\theta = \frac{\pi}{2}$: $\mathbf{R}_{\pi/2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \mathbf{J}$, the quarter-turn of §19.1.
- $\theta = \pi$: $\mathbf{R}_\pi = -\mathbf{I}$, the half-turn.
- $\theta = \frac{\pi}{4}$: $\mathbf{R}_{\pi/4} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, and it rotates $[\,2, 1\,]$ to $\frac{1}{\sqrt{2}}[\,1, 3\,]$. Both have length $\sqrt{5}$.

24.4. Every point turns, and the angle-addition formulas

The matrix was built from two points. Linearity guarantees it rotates every other point too, and it is instructive to watch that happen on the unit circle:

$$\mathbf{R}_\theta \begin{bmatrix} \cos t \\ \sin t \end{bmatrix} = \begin{bmatrix} \cos\theta\cos t - \sin\theta\sin t \\ \sin\theta\cos t + \cos\theta\sin t \end{bmatrix}.$$

Rotation should produce $[\,\cos(t+\theta), \sin(t+\theta)\,]$, and so the two entries are the **angle-addition formulas**:

$$\cos(t+\theta) = \cos t\cos\theta - \sin t\sin\theta, \qquad \sin(t+\theta) = \sin t\cos\theta + \cos t\sin\theta.$$

They are one statement about one matrix. Anyone who remembers $\mathbf{R}_\theta$ can recover them by multiplying it into $[\,\cos t, \sin t\,]$.

24.5. $\mathbf{R}_\theta$ is orthogonal, with determinant 1

Theorem 24.1. *For every angle θ,*

$$\mathbf{R}_\theta^{\mathsf{T}}\mathbf{R}_\theta = \mathbf{I} \qquad \textit{and} \qquad \det\mathbf{R}_\theta = 1.$$

Proof. Multiply out, using $\cos^2\theta + \sin^2\theta = 1$:

$$\begin{aligned} \mathbf{R}_\theta^{\mathsf{T}}\mathbf{R}_\theta &= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\theta + \sin^2\theta & -\cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta + \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{bmatrix} = \mathbf{I}. \end{aligned}$$

And $\det \mathbf{R}_\theta = \cos\theta \cdot \cos\theta - (-\sin\theta)\sin\theta = \cos^2\theta + \sin^2\theta = 1$. □

So every rotation preserves lengths and angles (§20.5) and orientation and area ($\det = 1$). And $\mathbf{R}_\theta$ is exactly the first family of §21.2, $\begin{bmatrix} a & -c \\ c & a \end{bmatrix}$, with $a = \cos\theta$ and $c = \sin\theta$. Conversely every member of that family is a rotation, because a point $[\,a, c\,]$ on the unit circle is $[\,\cos\theta, \sin\theta\,]$ for some θ. In the plane:

$$\text{orthogonal with } \det = +1 \quad \Longleftrightarrow \quad \text{a rotation } \mathbf{R}_\theta.$$

24.6. Composing and undoing rotations

Rotating by β and then by α is, by §15.4, the product $\mathbf{R}_\alpha \mathbf{R}_\beta$, and geometrically it is a rotation by $\alpha + \beta$. The angle-addition formulas confirm it entry by entry:

$$\mathbf{R}_\alpha \mathbf{R}_\beta = \begin{bmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -(\cos\alpha\sin\beta + \sin\alpha\cos\beta) \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & \cos\alpha\cos\beta - \sin\alpha\sin\beta \end{bmatrix} = \mathbf{R}_{\alpha+\beta}.$$

Since $\alpha + \beta = \beta + \alpha$, it follows that $\mathbf{R}_\alpha \mathbf{R}_\beta = \mathbf{R}_\beta \mathbf{R}_\alpha$: rotations of the plane commute, even though matrices in general do not.

To undo a rotation by θ, rotate by $-\theta$. Since cosine is even and sine is odd,

$$\mathbf{R}_{-\theta} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \mathbf{R}_\theta^{\mathsf{T}}, \qquad \mathbf{R}_\theta^{\mathsf{T}} \mathbf{R}_\theta = \mathbf{R}_{-\theta}\mathbf{R}_\theta = \mathbf{R}_0 = \mathbf{I},$$

which is Theorem 24.1 again, read geometrically: *the transpose of a rotation is the rotation back.*

24.7. Reflections, by angle

The second family of §21.2, with $a = \cos\theta$ and $c = \sin\theta$, is

$$\mathbf{F}_\theta = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} = \mathbf{R}_\theta \ \mathrm{diag}(1, -1).$$

So $\mathbf{F}_\theta$ means "reflect across the x-axis, then rotate by θ". §21.3 showed it fixes $[\,1 + \cos\theta, \sin\theta\,]$. With the half-angle identities $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$, that vector is $2\cos\frac{\theta}{2}\,[\,\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\,]$. So $\mathbf{F}_\theta$ is the reflection across the line through the origin at angle $\theta/2$. Check: $\mathbf{F}_0 = \mathrm{diag}(1, -1)$ reflects across the x-axis (angle 0), and $\mathbf{F}_{\pi/2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ across $y = x$ (angle $\pi/4$).

Remark (Complex numbers). Multiplying $x + iy$ by $\cos\theta + i\sin\theta$ gives $(x\cos\theta - y\sin\theta) + i(x\sin\theta + y\cos\theta)$, the same arithmetic as $\mathbf{R}_\theta[\,x, y\,]$. Multiplying by a complex number of modulus 1 is a rotation of the complex plane, and $\mathbf{R}_\theta$ is its matrix. The rule $\mathbf{R}_\alpha\mathbf{R}_\beta = \mathbf{R}_{\alpha+\beta}$ is the rule that the angles of complex numbers add when they are multiplied.

25. Rotations in $\mathbb{R}^3$

25.1. About the z-axis

In space, a rotation turns about an *axis*, a line through the origin that stays fixed. The simplest axis is the z-axis. A rotation about it by θ turns the xy-plane exactly as $\mathbf{R}_\theta$ turns the plane and leaves heights alone:

$$\mathbf{e}_1 \mapsto [\,\cos\theta,\ \sin\theta,\ 0\,], \qquad \mathbf{e}_2 \mapsto [\,-\sin\theta,\ \cos\theta,\ 0\,], \qquad \mathbf{e}_3 \mapsto \mathbf{e}_3,$$

so, by the recipe of §15.1 in three dimensions,

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_\theta & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}.$$

25.2. About the x- and y-axes

"Counterclockwise" about an axis means counterclockwise as seen from the positive end of the axis, looking back toward the origin. The three coordinate directions follow each other in the cyclic order $x \to y \to z \to x$. A rotation about the z-axis turns x toward y, one about the x-axis turns y toward z, and one about the y-axis turns z toward x.

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}.$$

The signs in $\mathbf{R}_y$ look reversed, and the cyclic order is why. Turning z toward x sends $\mathbf{e}_3 \mapsto [\sin\theta, 0, \cos\theta]$ and $\mathbf{e}_1 \mapsto [\cos\theta, 0, -\sin\theta]$. Written in the order (z, x) the block is an ordinary $\mathbf{R}_\theta$; listed in the order (x, z), its off-diagonal signs swap places. Check with $\theta = \frac{\pi}{2}$: $\mathbf{R}_y$ sends $\mathbf{e}_3$ to $\mathbf{e}_1$ and $\mathbf{e}_1$ to $-\mathbf{e}_3$, a quarter-turn from z toward x.

25.3. Rotation with respect to a plane

Each of these rotations has a plane that it turns and an axis it fixes: $\mathbf{R}_z$ turns the xy-plane and fixes the z-axis, and so on. The axis is the line orthogonal to the plane. That is the general picture.

> A rotation of $\mathbb{R}^3$ by θ turns one plane through the origin by θ, within itself, and leaves the line orthogonal to that plane fixed.

Concretely, let $\mathbf{u}, \mathbf{w}$ be orthonormal vectors spanning the plane and $\mathbf{n}$ a unit vector orthogonal to both. The rotation sends

$$\mathbf{u} \mapsto \cos\theta\,\mathbf{u} + \sin\theta\,\mathbf{w}, \qquad \mathbf{w} \mapsto -\sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{w}, \qquad \mathbf{n} \mapsto \mathbf{n}.$$

Measured in the directions $\mathbf{u}, \mathbf{w}, \mathbf{n}$ instead of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, it has exactly the block form of $\mathbf{R}_z$. Its matrix in the standard directions is found by a change of basis, which comes later in the course.

25.4. Properties

Each of $\mathbf{R}_x, \mathbf{R}_y, \mathbf{R}_z$ is orthogonal: its columns are unit vectors, and any two of them are orthogonal. For $\mathbf{R}_z$, the first two columns are the columns of $\mathbf{R}_\theta$ with a 0 appended, and the third is $\mathbf{e}_3$. So rotations of space preserve lengths and angles, as they must. Their determinants are all 1; determinants of 3×3 matrices are defined in Part 7, and §39.3 computes these.

25.5. Order matters in space

Rotations of the plane commute (§24.6). Rotations of space do not. Take two quarter-turns,

$$\mathbf{R}_z\left(\tfrac{\pi}{2}\right) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_x\left(\tfrac{\pi}{2}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

and compose them both ways:

$$\mathbf{R}_x\mathbf{R}_z = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \mathbf{R}_z\mathbf{R}_x = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Different. Follow $\mathbf{e}_1$: turning about z first sends it to $\mathbf{e}_2$, and then turning about x sends that to $\mathbf{e}_3$. Turning about x first leaves $\mathbf{e}_1$ alone, and then turning about z sends it to $\mathbf{e}_2$. Hold a book and do it. This is the warning of §8.4, that $\mathbf{AB}$ need not equal $\mathbf{BA}$, made physical.

Remark. In $\mathbb{R}^n$ the simplest rotations turn one coordinate plane, the (i,j)-plane, and fix every other coordinate. Their matrix is the identity except in rows and columns i and j, where the four entries of $\mathbf{R}_\theta$ sit. These are called *Givens rotations*, and numerical software uses them to introduce zeros into matrices one entry at a time.

25.6. Practice

P1. $\mathbf{A}$ and $\mathbf{B}$ are 2×2 with $\det \mathbf{A} = 2$ and $\det \mathbf{B} = -5$. Find $\det(3\mathbf{A})$, $\det(\mathbf{A}^\mathsf{T})$, $\det(\mathbf{AB})$, $\det(\mathbf{BA})$ and $\det(\mathbf{ABA}^\mathsf{T})$.

P2. Find the area of the parallelogram with sides $[\,3,1\,]$ and $[\,1,4\,]$, and of the triangle with corners $\mathbf{0}$, $[\,3,1\,]$, $[\,1,4\,]$.

P3. Verify by direct expansion that $(a^2+c^2)(b^2+d^2)-(ab+cd)^2 = (ad-bc)^2$ when $[\,a,c\,] = [\,1,2\,]$ and $[\,b,d\,] = [\,3,1\,]$, and compare with base times height computed from the angle.

P4. Write $\mathbf{R}_{\pi/3}$ and rotate $[\,2,0\,]$ by $\pi/3$. Without multiplying matrices, what is $(\mathbf{R}_{\pi/3})^3$?

P5. Write the matrix of the reflection across the line through the origin at angle $\pi/6$, and check that it fixes $[\cos\frac{\pi}{6}, \sin\frac{\pi}{6}\,]$.

P6. Compute $\mathbf{R}_y(\frac{\pi}{2})\mathbf{R}_z(\frac{\pi}{2})\mathbf{e}_1$ and $\mathbf{R}_z(\frac{\pi}{2})\mathbf{R}_y(\frac{\pi}{2})\mathbf{e}_1$.

Answers

P1. $3^2 \cdot 2 = 18$; 2; $2(-5) = -10$; -10; $2 \cdot (-5) \cdot 2 = -20$.

P2. $\det \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} = 11$, so the parallelogram has area 11 and the triangle, half of it, $\frac{11}{2}$.

P3. $(1+4)(9+1)-(3+2)^2 = 50-25 = 25$ and $(1-6)^2 = 25$. ✓ The angle has $\cos\theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}}$, so $\theta = \frac{\pi}{4}$, and base times height is $\sqrt{5} \cdot \sqrt{10} \cdot \frac{1}{\sqrt{2}} = 5 = |1-6|$. ✓

P4. $\mathbf{R}_{\pi/3} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$, and $[\,2,0\,] \mapsto [\,1,\sqrt{3}\,]$. Three rotations by $\frac{\pi}{3}$ make a rotation by π, so $(\mathbf{R}_{\pi/3})^3 = \mathbf{R}_\pi = -\mathbf{I}$.

P5. Angle $\frac{\pi}{6} = \frac{\theta}{2}$ means $\theta = \frac{\pi}{3}$: $\mathbf{F}_{\pi/3} = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$, and $\mathbf{F}_{\pi/3}[\,\frac{\sqrt{3}}{2}, \frac{1}{2}\,] = [\,\frac{\sqrt{3}}{4}+\frac{\sqrt{3}}{4},\ \frac{3}{4}-\frac{1}{4}\,] = [\,\frac{\sqrt{3}}{2}, \frac{1}{2}\,]$. ✓

P6. $\mathbf{R}_z(\frac{\pi}{2})\mathbf{e}_1 = \mathbf{e}_2$, which $\mathbf{R}_y$ fixes: the first is $\mathbf{e}_2$. $\mathbf{R}_y(\frac{\pi}{2})\mathbf{e}_1 = -\mathbf{e}_3$, which $\mathbf{R}_z$ fixes: the second is $-\mathbf{e}_3$. Different, as §25.5 said to expect.

Looking ahead. Rotations and reflections were the orthogonal matrices. Part 5 returns to the last motion of the gallery, the projection, and builds the matrix that projects onto *any* line from the projection formula of the vector unit. Projections are defined by the property that doing one twice is the same as doing it once.